

AVP Passthrough Observation Protocol

Camera-mediated viewing as a substrate discriminator | DMT Code research registry

PURPOSE

A camera sensor and a human retina render laser speckle differently. Speckle is formed at the imaging plane, so the grain pattern a sensor records is not the grain pattern an eye receives. Viewing the projected diffraction pattern through a passthrough headset such as Apple Vision Pro inserts a camera, a processing pipeline, and a display between the wall and the observer. Any reported structure must therefore survive, change, or vanish across that substrate boundary, and each outcome is data. Structure that appears identically in direct view and in passthrough points to the projected pattern itself. Structure present in direct view but absent in passthrough points to the observer's own visual system. This protocol standardizes that comparison.

PASSTHROUGH SAFETY, READ BEFORE THE APPARATUS SECTION

The headset views the projected pattern on the wall, exactly as an observer does. It is never placed in the beam path and the beam is never aimed at the headset. Direct laser exposure can permanently damage camera sensors, and a damaged passthrough feed endangers the wearer. Position the headset observer at the standard 1 to 3 meters from the surface, at an angle where no specular reflection reaches the headset cameras.

STANDARD APPARATUS, IDENTICAL TO THE FIELD SHEET, PLUS HEADSET

1. 650 nm red laser pointer (Class IIIa / 3R, under 5 mW) on an adjustable stand. Hands-free mounting required.
2. Diffraction grating in the beam path, close to the aperture, fanning the beam into the observation field.
3. Projection surface: flat, light-colored, matte wall or screen, 3 to 5 meters from the laser.
4. Passthrough-capable headset, fully charged, passthrough mode confirmed working in the darkened room before the session begins. Record device model and software version on page 2.
5. Room darkened. The pattern should be the brightest object in both the direct and the passthrough view.

COMPARISON PROCEDURE, A B A

1. Align the rig per the field sheet standard alignment procedure. Pattern spread 0.5 to 1.5 meters wide.
2. Segment A, direct view, minimum 5 minutes: observe without the headset. Note or sketch anything structured, with times.
3. Segment B, passthrough view, minimum 5 minutes: don the headset, same seat, same distance. Observe the same region of the pattern. Note whether each element from segment A is present, changed, or absent, and anything new that appears only in passthrough.
4. Segment A2, direct view again, minimum 5 minutes: remove the headset and re-check every element.
5. Do not confer between observers until all segments are complete. Complete pages 2 and 3 immediately.
6. Optional: capture a screen recording or photo through the headset if the device supports it, and note the capture times. A recording that shows the structure is a different class of evidence than one that does not, and both outcomes belong in the record.
7. Submit the completed record at dmtcode.com/registry and mark the session type as passthrough.

Adults 18+ only. Read the safety card on page 4, including the passthrough rules, before setup.

Passthrough Session Record

Complete during and immediately after the A B A sequence. One sheet per observer per session.

SESSION

Date _____ Start time / end time _____
Location type (home, studio, outdoor, other) _____ Observer ID or initials (optional) _____
Observers present (count) _____ Session number in your series _____
Session type: PASSTHROUGH _____ Segment lengths A / B / A2 (minutes) _____

APPARATUS CONFIGURATION

Laser model / source _____ Wavelength (nm) _____
Grating in beam path (type / lines per mm if known) _____ Laser-to-surface distance (m) _____
Projected pattern width on surface (m) _____ Observer-to-surface distance (m) _____
Projection surface (wall paint, screen, other) _____ Ambient light (fully dark, dim, other) _____
Corrective lenses worn (none, glasses, contacts) _____ Mount stable and hands-free (Y/N) _____

HEADSET CONFIGURATION

Device model _____ Software / OS version _____
Passthrough mode or setting used _____ Display brightness setting _____
Any lens inserts fitted (none / prescription) _____ Recording captured (Y/N, format) _____

SUBSTRATE COMPARISON, PER ELEMENT OBSERVED IN SEGMENT A

- | | | |
|---|---|--|
| ☐ Present unchanged in passthrough | ☐ Present but altered in passthrough | ☐ Absent in passthrough |
| ☐ New elements only in passthrough | ☐ Present again in segment A2 | ☐ Absent in segment A2 |
| ☐ Speckle texture differs across views | ☐ Latency or motion artifacts noted | ☐ No structure in any segment |

OBSERVATION NOTES, DESCRIPTION ONLY. STATE WHICH SEGMENT EACH NOTE BELONGS TO.

Substrate Comparison Sketch Record

Sketch each element twice: as seen in direct view, and as seen in passthrough. Same row, side by side.

DIRECT VIEW	PASSTHROUGH VIEW
Element 1, direct During observation ☐ From recall ☐	Element 1, passthrough During observation ☐ From recall ☐
Element 2, direct During observation ☐ From recall ☐	Element 2, passthrough During observation ☐ From recall ☐
Element 3, direct During observation ☐ From recall ☐	Element 3, passthrough During observation ☐ From recall ☐

Confidence that sketches reflect what was observed (low / moderate / high) and why _____

Laser Safety Card

Keep with the apparatus. Review before every session.

THE INSTRUMENT

This protocol uses a Class IIIa / 3R visible laser, under 5 mW output at 650 nm, compliant with US FDA CDRH requirements for consumer laser products. At this class, the blink reflex normally protects against momentary accidental exposure, but deliberate or prolonged direct viewing can cause permanent eye injury.

ABSOLUTE RULES

- Never look into the laser aperture, powered or not.
- Never point the beam at eyes, faces, people, or animals.
- Never view the beam or its reflections through magnifying optics, binoculars, cameras with optical viewfinders, or any lens system. Optics concentrate the beam and defeat the eye's natural protection.
- Never direct the beam at vehicles or aircraft. Doing so is a federal crime in the United States.
- Never leave the laser powered and unattended.
- Remove reflective objects (mirrors, glass, polished metal, dark screens) from the beam path before powering on.
- Adults 18+ operate the laser. Keep it out of the reach of children.

PASSTHROUGH RULES

- The headset views the projected pattern on the wall only. Never place the headset in the beam path and never aim the beam at the headset, powered or not. Direct exposure can permanently damage the cameras.
- Confirm passthrough works in the darkened room before the session. If the feed drops or blacks out, remove the headset before moving around the room with the laser powered.
- Seat the headset observer where no specular reflection from glass, metal, or screens reaches the cameras.

SESSION HYGIENE

Observers view the projected pattern on the surface, never the aperture and never specular reflections. Power the laser off between observation periods. Confirm stable mounting before each power-on so the beam cannot swing across the room if the stand is bumped. If any observer reports afterimages persisting more than a few minutes, end the session; persistent visual changes after any suspected direct exposure warrant a prompt eye examination.

CARE

Store the laser with batteries removed. Keep the grating clean and dry; handle it by the edges only. Inspect the stand and mount for looseness before each session. Replace any component that no longer holds the beam steady.